Gianluca Hotz

#sqlsat

PASS SQLSATURDAY

SQL Server Modernization

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Who am I?

Gianluca Hotz | @glhotz | ghotz@ugiss.org

Independent Consultant, Founder and Mentor SolidQ

- 20+ years on SQL Server (from 4.21 in 1996)
- Database modeling and development, sizing and administration, upgrade and migration, performance tuning

Interests

- Relational model, DBMS architecture, Security, High Availability and Disaster Recovery

Community

- 20 years Microsoft MVP SQL Server (from 1998)
- Founder and President UGISS
  - User Group Italiano SQL Server (PASS Chapter)

Modernization?

Low Complexity

- Version upgrades (e.g. 2008 R2 -> 2017)
- Edition upgrades/downgrades (e.g. Standard -> Enterprise)

Medium Complexity

- Virtualization
- Consolidation
- Adding HA/DR

High Complexity

- Re-Platforming (e.g. Oracle -> SQL Server or Azure SQL Database)
- Re-Architecting (e.g. SQL Server -> Azure SQL Datawarehouse or Data Lake)

Holistic Approach

Data Platform technologies

Project types

- Hardware modernization and/or consolidation
- Virtualization (Private and Public Cloud)
- Software solution deployments
  - SharePoint, Navision, CRM, …

Infrastructure

- hardware, virtualization platforms, storage solutions, HA/DR & backup solutions…

Application development

Licensing

SQL Server flavors

On-Premises

SQL Server (Database Engine, SSAS, SSRS, SSIS, DQS, MDS, MLS)
Analytics Platform System (APS was SQL Server PDW)

IaaS

SQL Server (Database Engine, SSAS, SSRS, SSIS, DQS, MDS, MLS)

PaaS

Azure SQL Database, Azure SQL Data Warehouse
Amazon RDS for SQL Server
Richer Data Platform offering (e.g. Data Lake, Data Factory, etc.)

Changing platform

Can be easy

- Migrating databases to PaaS or containerized versions (but same RDBMS)
- or…

Hard: converting databases to other RDBMSs

- data must be physically migrated (different types, different representations,…)
- database code must be converted (constraints, triggers, functions, procedures,…)

Application code can be

- converted completely, to a new platform
- just adapted to run against the new RDBMS
- adapted to support multiple RDBMSs

Different kind of projects

Migration

- one time conversion
- solution switch-over at a certain point

Porting

- side-by-side conversion & evolution
- no switch-over, current solutions remain

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Migration Projects

Usually done by final customers (i.e. not ISVs)

Driving factors

- cost reduction
- move away from vendor(s)
- solution obsolescence
  - unreliability and/or supportability problems

Application

- may be converted to a new platform
- may be adapted to run with new RDBMS

Porting Projects

Usually done by ISVs

Driving factors

- get into new markets
  - cost reduction for customers (TCO)
  - more choice for customers
- become less dependent from vendor(s)

Application

- may be ported to new platform
  - happens rarely, means re-write or 2 codebases
- may be adapted to run with new RDBMS

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Different kind of Databases to migrate/port

Simple schema

- all the logic is inside the application
- simple access methods (mostly by cursors)
- very limited use of SQL dialects
- database is used mostly for storage

Complex schema

- much of the logic is inside the database
- heavy usage of SQL dialects
- database used as a rule and inference engine

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Origin of problems

Sometimes porting issues arise from

- misusage of functionalities
- bad decoupling of application/database code
- too much row oriented code in the database
- low technical skills of coders/architects

Never judge

- every application has it own story
- a lot of decisions may be compromises
- a lot of money was probably already invested

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Application portability

Code in the RDBMS

- has been a best practice for years
- still a very good solution for both
  - integrity
  - scalability
- however… portability is an issue!
  - different SQL dialects
  - different type systems
  - different engine features

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Changing Architecture

Can be much more harder!

More complex

- Low familiarity with non RBMS technologies
- Low familiarity with unstructured dta
- New implementation patterns
- ML & AI will stop to be buzzwords…

Version Upgrades: motivations

Compliancy (e.g. end of support and/or security)

Better Performance

New functionalities

Winter is coming!

Version Upgrades: barriers

Licensing Costs

Regressions risk

Application changes costs

Complexity

- Testing changes (application or certification processes)
- Planning for low downtime

Lack of time/resources/competencies

Edition upgrades/downgrades

Upgrades

- Motivations similar to version upgrades
- Barriers similar to version upgrades
  - Complexity usually lower because setup covers the scenario

Downgrades

- Main motivation to save money
- Barriers similar to version upgrades
  - Needs reinstallation, less impact for non production environment

Hardest problems

Factors that require application changes

- Breaking changes
- Discontinued functionalities

Plan affecting changes may cause performance regressions

- New Cardinality Estimator
- Hotfixes
- New QP strategies/behavior (e.g. Intelligent Query Processing)

Testing changes is hard!

- Complete Workload coverage
- Concurrent access

Database Compatibility Level

Sets behavior compatible with specified version of SQL Server

Version	Native Level	Supported Levels
SQL Server 2019	150	150, 140, 130, 120, 110, 100
SQL Server 2017	140	140, 130, 120, 110, 100
Azure SQL Database	130	140, 130, 120, 110, 100
SQL Server 2016	130	130, 120, 110, 100
SQL Server 2014	120	120, 110, 100
SQL Server 2012	110	110, 100, 90
SQL Server 2008 R2	100	100, 90, 80
SQL Server 2008	100	100, 90, 80
SQL Server 2005	90	90, 80
SQL Server 2000	80	80

DB Compatibility Level based certification

- Target for a database compatibility level
  - Stop certifying for any given platform (On-Premises, Cloud, PaaS/IaaS)
  - Stop certifying for a named SQL Server version
- Microsoft Database Compatibility Level Protection
  - Full functional protection (once assessment tools runs with no errors)
  - Query Plan shape protection (on comparable hardware)
  - Maintaining backward compatibility very important for SQL Server team…

DB Compatibility Level certification benefits

Simplified certification across platforms

- E.g. SQL Server On-Premises and Azure SQL Database Managed Instances

Improved risk management

- Decoupling application upgrade from database support upgrade cycles

Allow customers to upgrade/migrate to new/different versions

- Benefits from new features Without changing functionalities
- Not everything is gated by Database Compatibility levels e.g.
  - Adaptive Joins: gated
  - Auto-tuning, new Query memory gateways, leveraging NUMA architecture, Columnstore indexes: non-gated

Functional Change Protection

Breaking Changes

- Behavior changes resulting in different outcomes!
- Not all protected under Database Compatibility

Protected changes examples

- Improved accuracy in compatibility level 130+
- Conversion behavior setting LANGUAGE context info
- Conversion behavior parsing dates

Non protected changes examples

- Column names changes in system objects

Demo

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Functional Change Protection & Trace Flags

Sometimes functional changes are gated by Trace Flags

Example: truncation error 8152

- Gated by trace flag 460
- Will be on by default in 2019 with Compatibility Level 150
- https://blogs.msdn.microsoft.com/sql_server_team/string-or-binary-data-would-be-truncated-replacing-the-infamous-error-8152

Discontinued functionalities

Not protected by Compatibility Level!

- Removed from database engine
- Need to change queries using it

Examples

- **FASTFIRSTROW**
  - replaced by **OPTION (FAST n)** in SQL Server 2012
- **sp_dboption**
  - system stored procedure **removed** in SQL Server 2012

Query Plan shape protection

Plan-affecting changes

- Query Optimizer fixes (Trace Flag 4199)
- Changes to Cardinality Estimator

These will very likely cause regressions!

- Especially Cardinality Estimator
- Many queries better performance, other worse
- Are you sure you understood how fixes works?
- What about fixes for incorrect result causing regressions?

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Query Optimizer Fixes with TF 4199

SQL Server 2016+ QO fixes from previous versions **enabled by default** under the **latest** level

- previously you had to enable them but you enabled also current version, post-RTM, fixes
- **zero reported errors in support cases after changing the default**

Fully Documented

- https://support.microsoft.com/en-us/help/974006/sql-server-query-optimizer-hotfix-trace-flag-4199-servicing-model

Compatibility Level	TF 4199	QO changes from previous level	QO changes for current version post-RTM
100-120	**Off**	**Disabled**	Disabled
	On	Enabled	Enabled
130	**Off**	**Enabled**	Disabled
	On	Enabled	Enabled
140	**Off**	**Enabled**	Disabled
	On	Enabled	Enabled

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Query Optimizer fixes further control

SQL Server 2016+ database-level granularity

ALTER DATABASE SCOPED CONFIGURATION SET **QUERY_OPTIMIZER_HOTFIXES** = ON|OFF

SQL Server 2016 SP1+ query-level granularity

OPTION(USE HINT('**ENABLE_QUERY_OPTIMIZER_HOTFIXES**'))

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Cardinality Estimator problems

Essential for query plan generation

Was largely based on SQL Server 7.0

Redesigned in SQL Server 2014+

- Ascending data estimation, correlation assumptions, join containment, fixed estimation values (e.g. MTVF)
- Will most probably cause regressions in performance!

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Controlling Cardinality Estimation

SQL Server 2014

- Bound to Database Compatibility Level 😬
- Trace Flags (global, session and query levels)
  - Trace flag 2312 reverts to CE version 70
  - Trace flag 9481 use CE version 120

SQL Server 2016+ database-level configuration

- ALTER DATABASE SCOPED CONFIGURATION SET **LEGACY_CARDINALITY_ESTIMATION** = ON|OFF

SQL Server 2016 SP1+ query-level configuration

- OPTION(USE HINT('**FORCE_LEGACY_CARDINALITY_ESTIMATION**'))
- OPTION(USE HINT('**FORCE_DEFAULT_CARDINALITY_ESTIMATION**'))

Query Optimizer Compatibility Level

SQL Server 2017 CU10 introduces query-level granularity

- OPTION(USE HINT('**QUERY_OPTIMIZER_COMPATIBILITY_LEVEL_n**'))
- Includes QO fixes enabled by default up to that level
- TF 4199 enabled globally will still enable all fixes

https://blogs.msdn.microsoft.com/sql_server_team/developers-choice-hinting-query-execution-model

Query Optimizer Hints

SQL Server 2016 SP1+

Control QO behavior

Mainly to replace trace flags

USE HINT	Trace Flag	DB Option
DISABLE_OPTIMIZED_NESTED_LOOP	2340	
FORCE_LEGACY_CARDINALITY_ESTIMATION	9841	Yes
ENABLE_QUERY_OPTIMIZER_HOTFIXES	4199	Yes
DISABLE_PARAMETER_SNIFFING	4136	Yes
ASSUME_MIN_SELECTIVITY_FOR_FILTER_ESTIMATES	4137 (Old CE) 9471 (New CE)	
DISABLE_OPTIMIZER_ROWGOAL	4138	
ENABLE_HIST_AMENDMENT_FOR_ASC_KEYS	4139	
ASSUME_JOIN_PREDICATE_DEPENDS_ON_FILTERS	9476 (New CE)	
FORCE_DEFAULT_CARDINALITY_ESTIMATION	2312	
QUERY_OPTIMIZER_COMPATIBILITY_LEVEL_n		
QUERY_PLAN_PROFILE		

https://blogs.msdn.microsoft.com/sql_server_team/developers-choice-use-hint-query-hints

Microsoft recommended upgrade Plan

Upgrade to **latest** SQL Server version and keeping **source** compatibility level

→

Enable Query Store

→

Create a baseline of the workload

→

Move to latest compatibility level

→

Quickly fix regressions by forcing last known good plan!

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Upgrade Strategies

Side-by-side

Allow OS Upgrade

Easier testing

Easier rollback strategy

Less downtime

In-Place

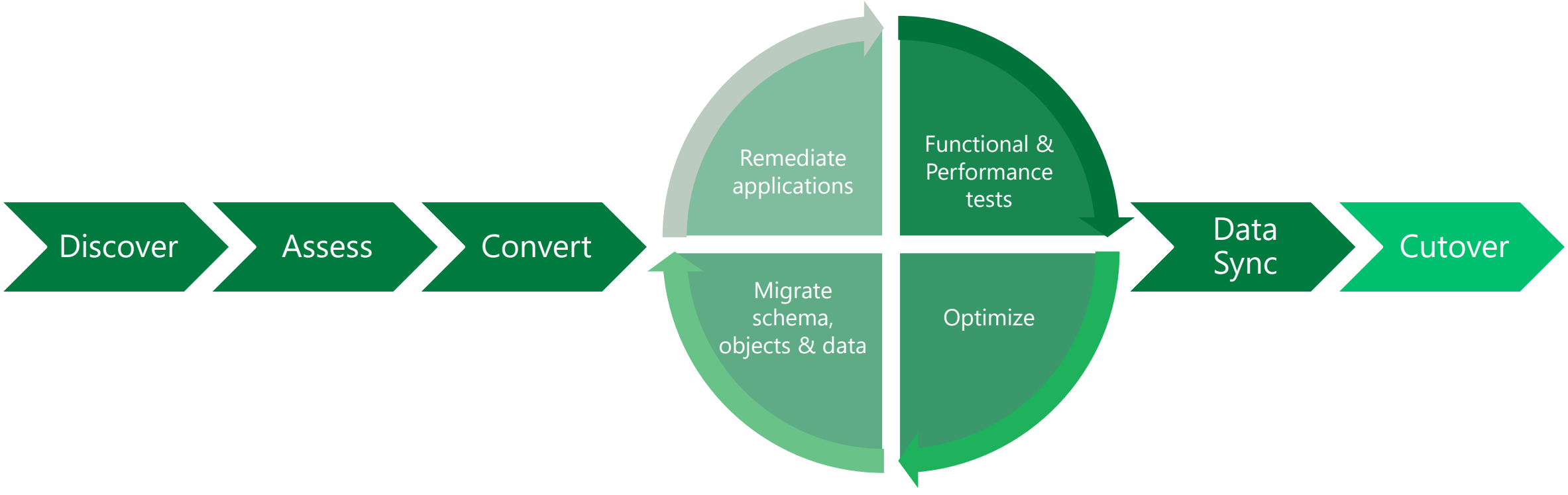
No additional Hardware

No data migration

Side-by-side Upgrade Service Availability

Synchronization Strategy	Notes
Log Shipping	Cutover typically in minutes
Replication	Cutover potentially in seconds
Backup/Restore	From minutes to hours (depends on strategy e.g. usage of differential and/or log backups)
Filesystem/SAN Copy	Dependent on technology
Database Mirroring/Availability Groups	Cutover typically in seconds

Upgrade process to minimize risks



Discover Phase

- Which SQL Server versions do I have?
- Which Editions am I running?
- Which SQL Server components are installed?
- How many cores are on each server?
- How many databases are in each instance?
- What are the sizes of all my databases?
- What are the settings for each instance and database?

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Assessment Tools

Microsoft Assessment and Planning (MAP) Toolkit

SQL PowerDoc

- PowerShell Script gather data from WMI in XML
- Windows and SQL Server rich inventory in Excel
- Best Practices Reports in Excel
- Consolidates multiple server/instances
- Official https://github.com/kendalvandyke/sqlpowerdoc
- Newer https://github.com/bryan5989/SQLPowerDoc

Quest Enterprise Reporter for SQL Server

- Based on previously free Discovery Wizard for SQL Server
- Part of a suite for compliancy covering several other products (e.g. AD)

Microsoft Upgrade/Migration Tools

Upgrade Advisor

Data Migration Assistant (DMA)

Data Experimentation Assistant (DEA)

Data Migration Service (DMS)

SQL Server Migration Assistant (SSMA)

SQL Server Data Tools for Visual Studio (SSDT)

SQL Server Management Studio (SSMS)

- “Export Data Tier Application”
- “Deploy Database to Microsoft Azure SQL Database” Wizard

Upgrade Advisor

Static Analysis

- Database Engine
- Analysis Services
- Reporting Services
- Integration Services

Dynamic Analysis

- Trace Files

Obsolete!

- Use only if source database non supported
- You can still find it in SQL Server Feature Packs (2014 latest)

Data Migration Assistant

On-premises instances assessment

- Blocking issues
- Partially supported or unsupported features

Discover issues that may affect compatibility

- Breaking Changes, Behavior Changes, Deprecated Features

Discover new features that may benefit after upgrading

- Performance, security, Storage

Azure SQL Database SKU Recommendations

- Performance counters analysis to suggest SKU Azure SQL Database

Performs the migration

Versions supported by DMA

Source

SQL Server 2005

SQL Server 2008

SQL Server 2008 R2

SQL Server 2012

SQL Server 2014

SQL Server 2016

SQL Server 2017 on Windows

Destination

SQL Server 2012

SQL Server 2014

SQL Server 2016

SQL Server 2017
on Windows and Linux

Azure SQL Database

Azure SQL Database
Managed Instance

Bulk Analysis

Repository instances/databases to assess in SQL/CSV

Automation with PowerShell

Import in reporting database

Reports with PowerBI

DMA Problems

- Cannot save sessions!
  - Only Assessment data in JSON format
- Azure SQL Database SKU Recommendations
  - Command Line Interface only
  - Currently not working…
- Bulk Analysis
  - Cumbersome process
  - Errors
    - dmaDataCollector crashes with pubs, dmacmd.exe is a closed project
    - LoadWarehouse.sql needs to be fixed

Data Migration Assistant Resources

Docs

https://docs.microsoft.com/en-us/sql/dma/dma-overview

https://docs.microsoft.com/en-us/sql/dma/dma-consolidatereports

Blog

https://blogs.msdn.microsoft.com/datamigration/tag/dma

https://blogs.msdn.microsoft.com/datamigration/tag/data-migration-assistant

Database Experimentation Assistant

A/B testing solution for SQL Server

- Workload comparison
- e.g. upgrades, new indexes, etc.

Capture

- SQL Trace
- Extended Events (V2.6+)

Replay

- DReplay integration

Analysis

- Performance improvements and regressions

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DEA Analysis Example

Database Experimentation Assistant

Analysis Reports

Connected Report Server: DEAGAWSVR2016 Switch Server

NEW REPORT

STATUS	NAME	DATE
✓	Test08To14Smu	2018/07/18
✓	ADWTPS100	2018/07/18
✓	2016Migration	2018/03/13

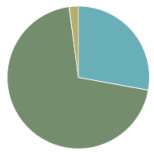
2016Migration

Threshold 5% Export Print

Target	Instance Name	Product Name	Trace File
Target 1	SQL2008SOURCE	SQL Server 2008 SP4 MS15-059	C:\Users\mojain\Documents\sql20008-small\Trace.trc
Target 2	SQL2016TRACE	SQL Server 2016	C:\Users\mojain\Documents\sql2016-small\Trace.trc

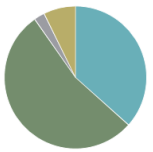
QUERY DISTRIBUTION

By execution count



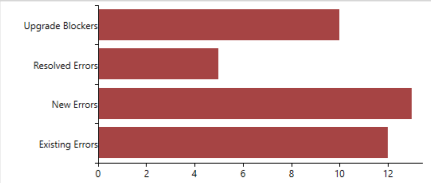
Degraded [14701] 27.9%
Improved [36951] 70%
Cannot Evaluate [1] 0%
Same [1108] 2.1%

By distinct queries



Degraded [15] 36.6%
Improved [22] 53.7%
Cannot Evaluate [1] 2.4%
Same [3] 7.3%

Distinct error queries count by type



Error ID count

Error ID 156 (11) 36.7%
Error ID 208 (6) 20%
Error ID 2714 (5) 16.7%
Error ID 3701 (5) 16.7%
Error ID 539 (3) 10%

Top queries in workload

Top Improved Queries | Top Degraded Queries

Hash Id	Query Text	Mean Duration on Target 1 (µs)	Mean Duration on Target 2 (µs)	Duration Difference (µs)
-1616155278339986948	IF OBJECT_ID ((STR), (STR)) IS NOT NULL DROP TABLE DBO.EMPLOYEEONE;	102353	9105	-93248
-1704013646769530258	SELECT DISTINCT PP.LASTNAME, PP.FIRSTNAME FROM PERSON.PERSON PP JOIN	181508	127802	-53706

Database Experimentation Assistant

Analysis Reports

Connected Report Server: DEAGAWSVR2016 Switch Server

NEW REPORT

STATUS	NAME	DATE
✓	Test08To14Smu	2018/07/18
✓	ADWTPS100	2018/07/18
✓	2016Migration	2018/03/13

2016Migration > Error

Threshold 5% Export Print

Target	Instance Name	Product Name	Trace File
Target 1	SQL2008SOURCE	SQL Server 2008 SP4 MS15-059	C:\Users\mojain\Documents\sql20008-small\Trace.trc
Target 2	SQL2016TRACE	SQL Server 2016	C:\Users\mojain\Documents\sql2016-small\Trace.trc

ERROR QUERIES

Error Type	Query Count	Execution Count
Existing Errors (Errors on target 1 server that continue to exist on the target 2 server.)	12	1128
New Errors (Errors which are new on the target 2 server.)	13	3552
Resolved Errors (Errors which existed on target 1 server but resolved in target 2 server.)	5	14
Upgrade Blockers (Errors that will block upgrade to target server.)	10	3519

New Errors | Existing Errors | Resolved Errors | Upgrade Blockers

Query Text	Error ID	Error Count	Error Text
SELECT PRODUCTID, LINETOTAL FROM SALES.SALESORDERDETAIL WHERE UNITPRICE < {##},{##} ORDER BY PRODUCTID, LINETOTAL COMPUTE SUM(LINETOTAL) BY PRODUCTID;	156	372	Incorrect syntax near the keyword 'COMPUTE'.
SELECT PRODUCTID, ORDERQTY, SUM(LINETOTAL) AS TOTAL FROM SALES.SALESORDERDETAIL WHERE UNITPRICE < {##},{##} GROUP BY PRODUCTID, ORDERQTY ORDER BY PRODUCTID, ORDERQTY COMPUTE SUM(SUM(LINETOTAL)) BY PRODUCTID, ORDERQTY COMPUTE SUM(SUM(LINETOTAL));	156	397	Incorrect syntax near the keyword 'COMPUTE'.
SELECT SALESPERSONID, CUSTOMERID, ORDERDATE, SUBTOTAL, TOTALDUE FROM SALES.SALESORDERHEADER ORDER BY SALESPERSONID, ORDERDATE COMPUTE SUM(SUBTOTAL), SUM (TOTALDUE) BY SALESPERSONID;	156	342	Incorrect syntax near the keyword 'COMPUTE'.
SELECT PRODUCTID, ORDERQTY, LINETOTAL FROM SALES.SALESORDERDETAIL COMPUTE SUM (ORDERQTY), SUM(LINETOTAL);	156	365	Incorrect syntax near the keyword 'COMPUTE'.
SELECT PRODUCTID, ORDERQTY, UNITPRICE, LINETOTAL FROM SALES.SALESORDERDETAIL WHERE UNITPRICE < {##},{##} COMPUTE SUM(ORDERQTY), SUM(LINETOTAL);	156	327	Incorrect syntax near the keyword 'COMPUTE'.
SELECT PRODUCTID, ORDERQTY, UNITPRICE, LINETOTAL FROM SALES.SALESORDERDETAIL WHERE UNITPRICE < {##},{##} ORDER BY PRODUCTID, ORDERQTY, LINETOTAL COMPUTE SUM(LINETOTAL) BY PRODUCTID, ORDERQTY COMPUTE SUM(LINETOTAL) BY PRODUCTID;	156	354	Incorrect syntax near the keyword 'COMPUTE'.
SELECT PRODUCTID, LINETOTAL FROM SALES.SALESORDERDETAIL WHERE UNITPRICE < {##},{##} ORDER BY PRODUCTID, LINETOTAL COMPUTE SUM(LINETOTAL), MAX(LINETOTAL) BY PRODUCTID;	156	375	Incorrect syntax near the keyword 'COMPUTE'.
SELECT PRODUCTID, ORDERQTY, UNITPRICE, LINETOTAL FROM SALES.SALESORDERDETAIL WHERE UNITPRICE < {##},{##} ORDER BY PRODUCTID COMPUTE SUM(ORDERQTY), SUM(LINETOTAL) BY PRODUCTID COMPUTE SUM(ORDERQTY), SUM(LINETOTAL);	156	323	Incorrect syntax near the keyword 'COMPUTE'.
SELECT CUSTOMERID, ORDERDATE, SUBTOTAL, TOTALDUE FROM SALES.SALESORDERHEADER WHERE	156	331	Incorrect syntax near the keyword 'COMPUTE'.

Versions supported by DEA

Source

SQL Server 2005+

Target

SQL Server 2005+

Analysis

SQL Server 2008+

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Data Migration Service Resources

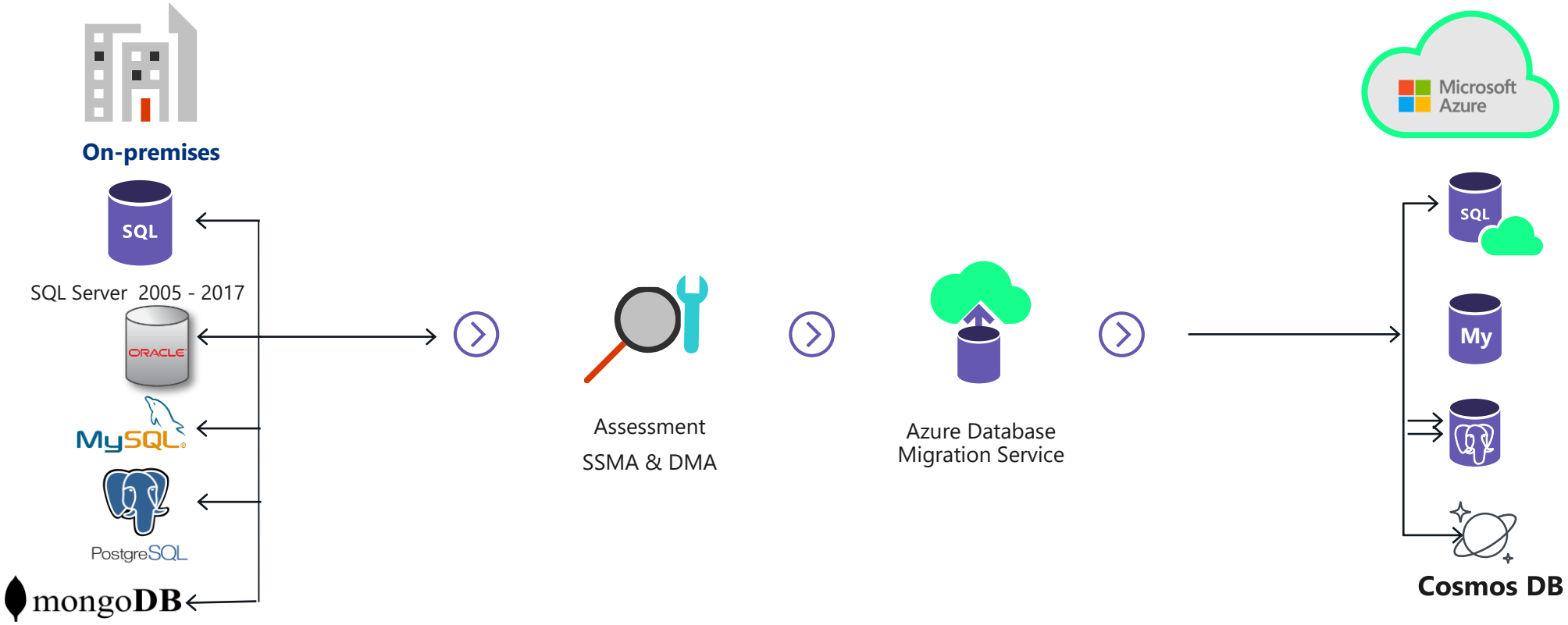
Docs

https://docs.microsoft.com/en-us/sql/dea/database-experimentation-assistant-overview

Blog

https://blogs.msdn.microsoft.com/datamigration/tag/dea

Azure Database Migration Service



DMS Migration Scenarios

SQL Server -> Azure SQL Database Offline

SQL Server -> Azure SQL Database Online

SQL Server -> Azure SQL Database Managed Instance Offline

SQL Server -> Azure SQL Database Managed Instance Online

MySQL -> Azure DB for MySQL Online

PostgreSQL -> Azure DB for PostgreSQL Online

Data Migration Service Resources

Docs

https://docs.microsoft.com/en-us/azure/dms/dms-overview

https://docs.microsoft.com/en-us/azure/dms/tutorial-sql-server-to-azure-sql

Blog

https://blogs.msdn.microsoft.com/datamigration/tag/azure-database-migration-service

https://blogs.msdn.microsoft.com/datamigration/tag/dms

SQL Server Migration Assistant

Migration effort assessment

Code conversion

Data migration

Testing

Deployment

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Platforms supported by SSMA

Source

Microsoft Access

DB2

MySQL

Oracle

SAP ASE (Sybase ASE)

Destination

SQL Server 2008

SQL Server 2008 R2

SQL Server 2012

SQL Server 2014

SQL Server 2016

SQL Server 2017 on Windows and Linux

SQL Server 2019 on Windows and Linux

Azure SQL Database

Azure SQL Database Managed Instance

Azure SQL Data Warehouse (Oracle only)

SSMA for Oracle: how it works (101)

- Creates two “helper” databases
  - sysdb
  - ssmatesterdb
- Stores metadata in the file system
  - data stored in xml compressed format
    - can rename to .gzip and uncompress it
  - shred xml into lot of files for caching
    - reports directory
      - can be safely deleted and re-generated
      - metadata reside in the original .mb files

SSMA for Oracle: sysdb helper database

Provides

- code infrastructure to emulate features
  - system functions
  - schema features (e.g. sequences)
  - engine features (e.g. autonomous transactions)
- session bound data storage
- emulated metadata storage (e.g. packages)

Maintenance

- protect metadata
- performance related depends on workload

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SSMA for Oracle: does it really work?

Yes, but don't expect miracles!

- Works well with simple schemas…
- Usually won't do all the work by itself

Very useful for

- assessing efforts
- assessing problems
- doing prototypes
- as a learning tool
- to translate/emulate some functionalities

Testing

Nothing can replace good testing!

- Hard to have tests with good coverage of the whole workload
- Sometimes is more cost effective to run fewer tests and fix the problems later…

SQL Server DReplay infrastructure

- Often impossible to capture and replay even a fraction of the workload

Workload Tools

Very good for side-by-side, real-time workload replay

- Captures XE batch completed and replay commands on-the-fly

Captures execution telemetry (cpu, read, writes, duration)

Reports to find regressions

Developed by a well-known MVP ☺

https://github.com/spaghettidba/WorkloadTools

DBA Tools

Lot of PowerShell cmdlets to make the DBA life easier

Many cmdlets target specifically migration/upgrade scenarios

- E.g copy logins with synch option

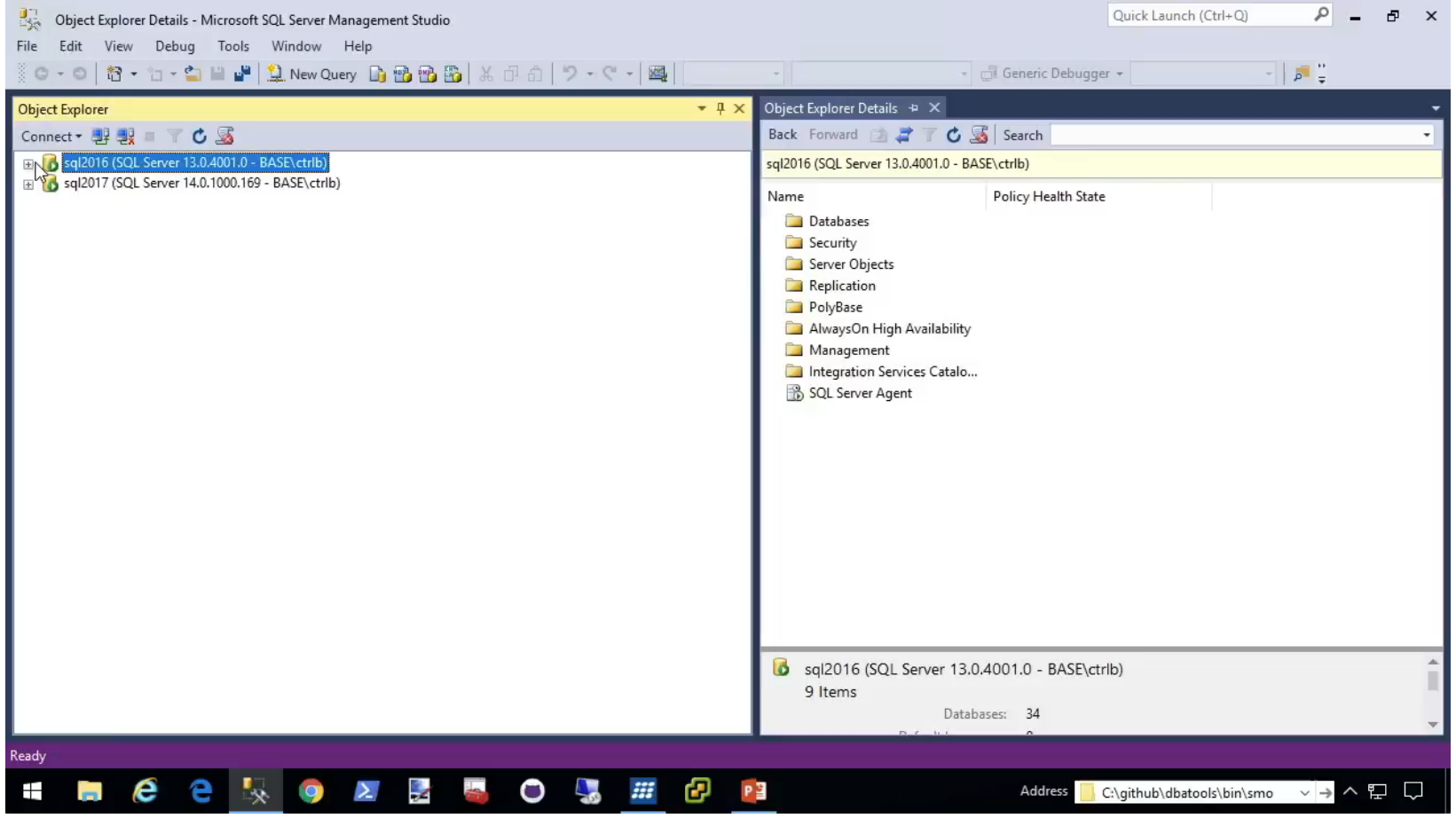
One stop cmdlet migrating an

- All instance-level assets (e.g. credentials, linked servers, etc.)
- Via backup/restore automation (just specify network share)

https://dbatools.io

Start-DbaMigration

https://youtu.be/hg8tovMRX2k



After the upgrade: remember the plan

Upgrade to **latest** SQL Server version and keeping **source** compatibility level

→

Enable Query Store

→

Create a baseline of the workload

→

Move to latest compatibility level

→

Quickly fix regressions by forcing last known good plan!

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Automatic Tuning

Automatic plan correction

- Automatically force last good plan when regression detected
- Minimum 10 CPU seconds improvement
- **sys.dm_db_tuning_recommendations** for manual corrections

SQL Server 2017 and Azure SQL Database

https://docs.microsoft.com/en-us/sql/relational-databases/automatic-tuning/automatic-tuning

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Query Tuning Assistant

Well-defined workflow targeting patterns for query regressions

- Independence vs. Correlation
- Simple Containment vs. Base Containment
- Multi-statement table-valued function (MSTVF) fixed cardinality guess

SQL Server Management Studio Wizard-Based Tool

Can be automated via PowerShell

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Query Tuning Assistant plan

Upgrade to **latest** SQL Server version and keeping **source** compatibility level

Enable Query Store

Create a baseline of the workload

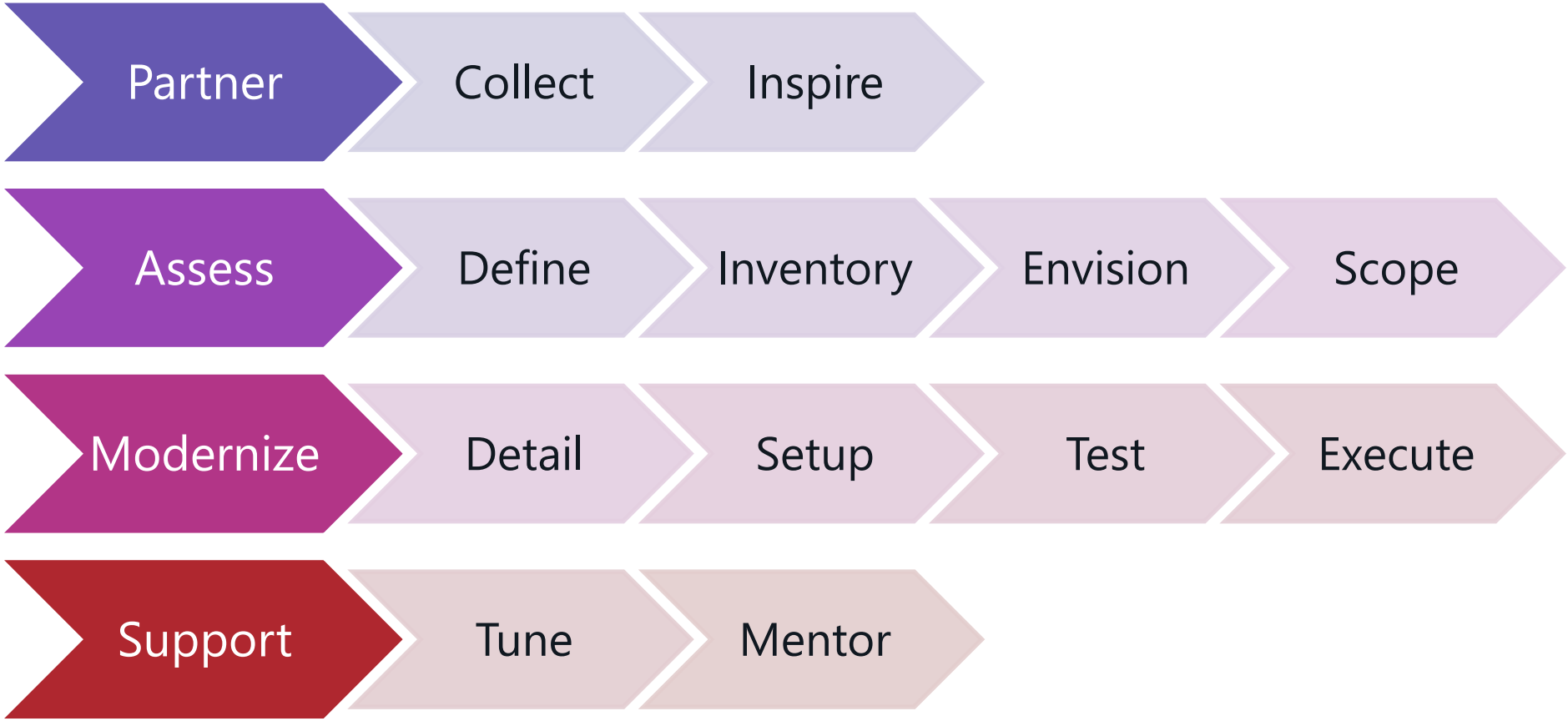
Move to latest compatibility level (or any target level to test)

Collect workload with new level and start comparing with baseline

Instead of only forcing last known good plan, experiment with tuning recommendation actions

https://docs.microsoft.com/en-us/sql/relational-databases/performance/upgrade-dbcompat-using-qta

Modernization Stream Phases in a SQL Project



Assessment phase

Define
- Finalize assessment's scope
- Finalize questionnaires to be sent to project representatives
- Activities kick-off

Inventory
- Detailed inventory of Windows Server & SQL Servers
- Static analysis of databases & SQL code
- Project representatives interviews (questionnaires)

Envision
- Consolidate & analyze inventory data
- Review configuration, performance, reliability and security best practices
- Prepare modernization scenarios and macro action-plan

Scope
- Discuss deliverables and scenarios with customer
- Finalize modernization scenarios
- Prepare next phases offering (Modernize & Support)

Assessment phase: reducing uncertainty

Better planning/flexibility

- Size better understood results in reduced expectations and scope
- Allows to run several Modernization phases at different times

Leaves the customer with something tangible

- Detailed assets inventory
- Performance/Security/Reliability automatic assessment
- Static analysis of databases code
- Possibility to rescope/change scenarios
- Enough data to independently run the project

On-Premises to Azure PaaS/IaaS migrations

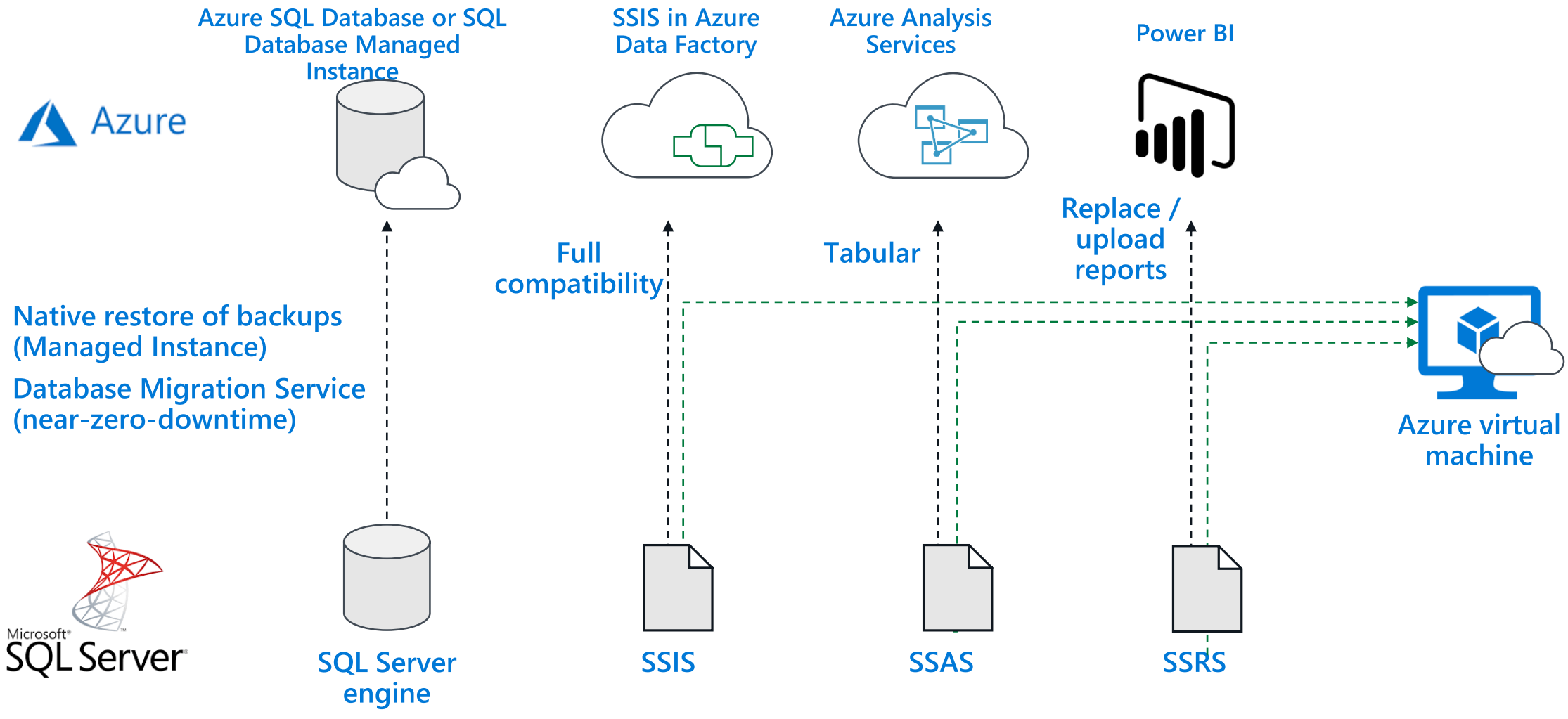
IaaS

- SQL Server on Azure VMs
- SQL Server on Azure Containers
- SQL Server on Azure Kubernetes

PaaS

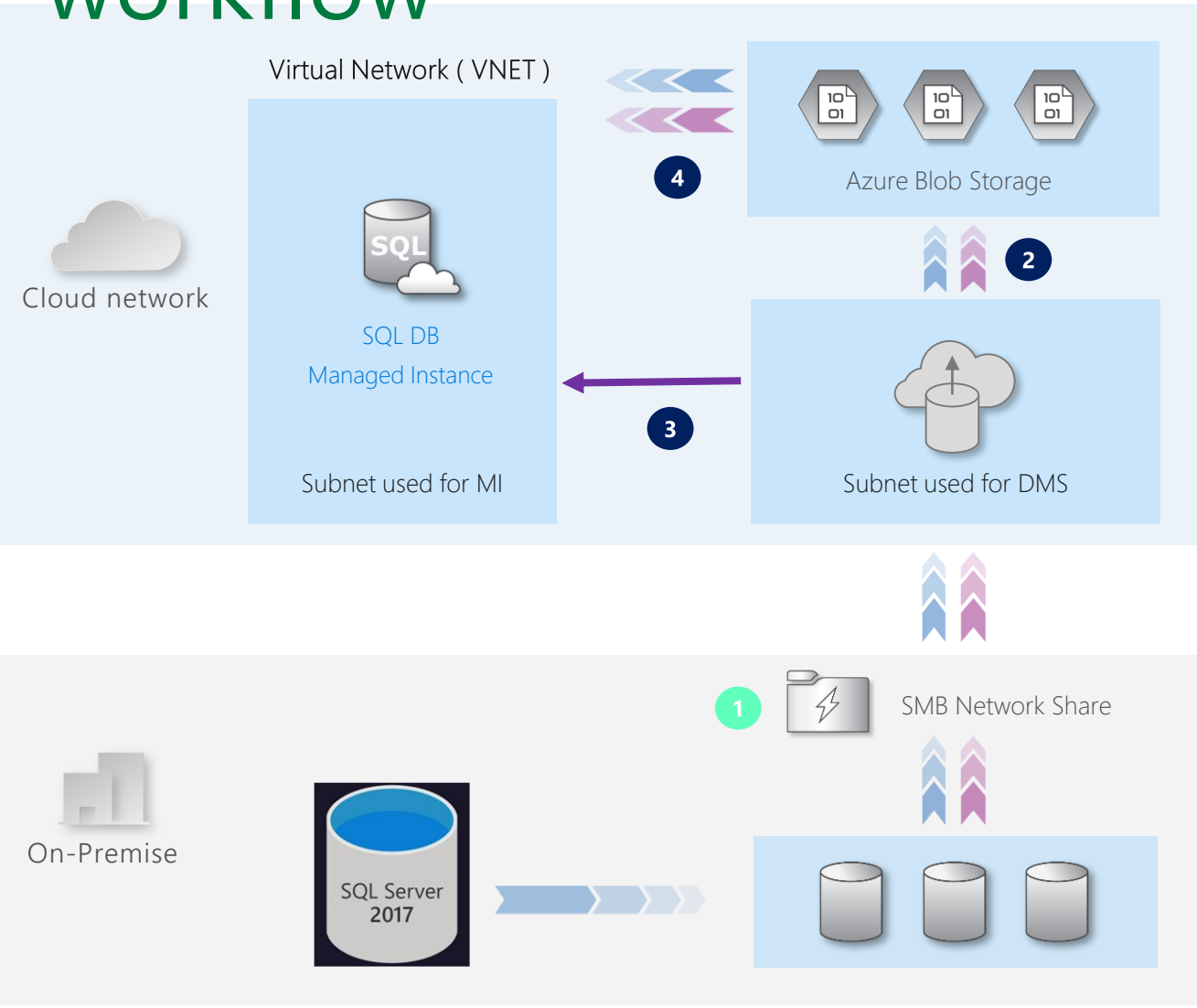
- Azure SQL Database
- Azure SQL Database Managed Instance

On-Premises to SQL Server PaaS migrations



- Native restore of backups (Managed Instance)
- Database Migration Service (near-zero-downtime)

On-Premise to Managed Instance online workflow



Legend

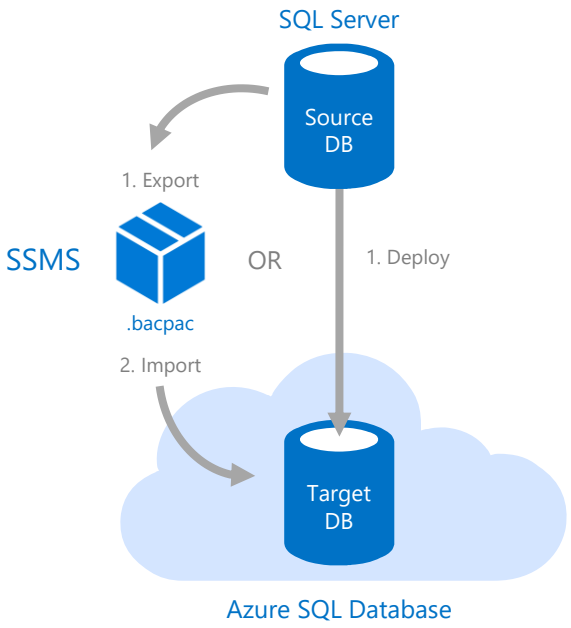
- Full Database backup files
- Transaction log backup files
- Site to site connectivity (VPN or ExpressRoute)

1. Provide existing backups in network share
2. DMS upload backup files to Azure storage
3. DMS initiate the migration to Azure SQL MI
4. Full backup restored and Transaction log backups continuously applied until cutover

Provide Tail-Log backup, initiate cutover in DMS and change the application connection strings

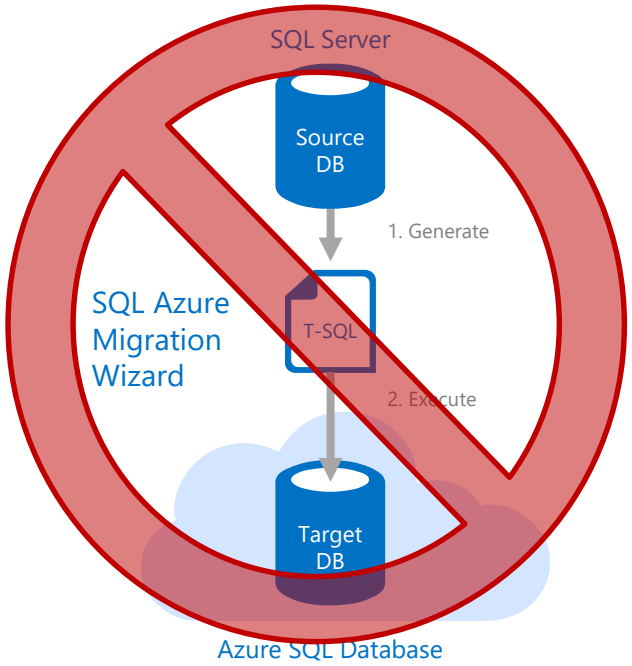
Other on-premises to PaaS strategies

Method 1



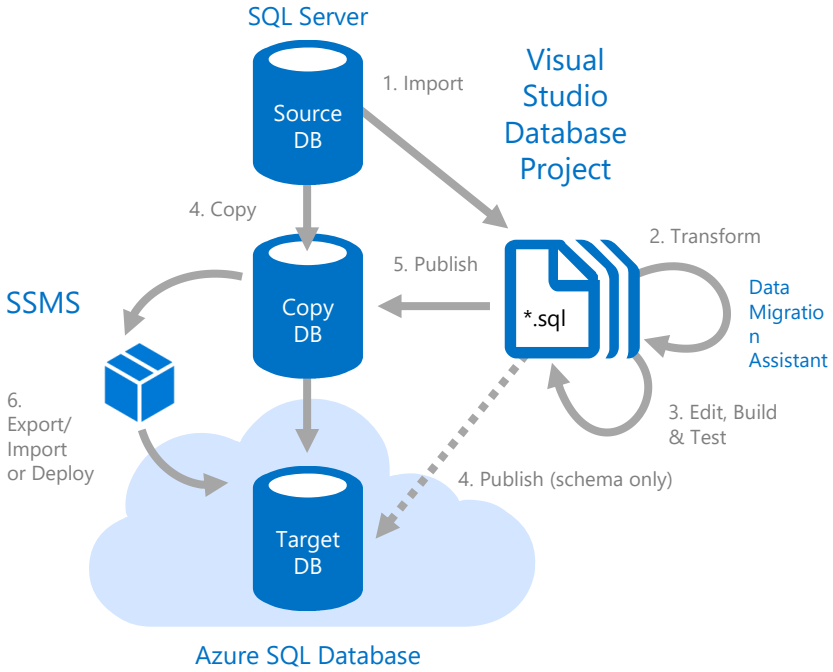
Compatible database: manual copy or deploy wizard in SSMS

Method 2



Database almost compatible: ~~SQL Azure Migration Wizard~~ DMA and SSMS

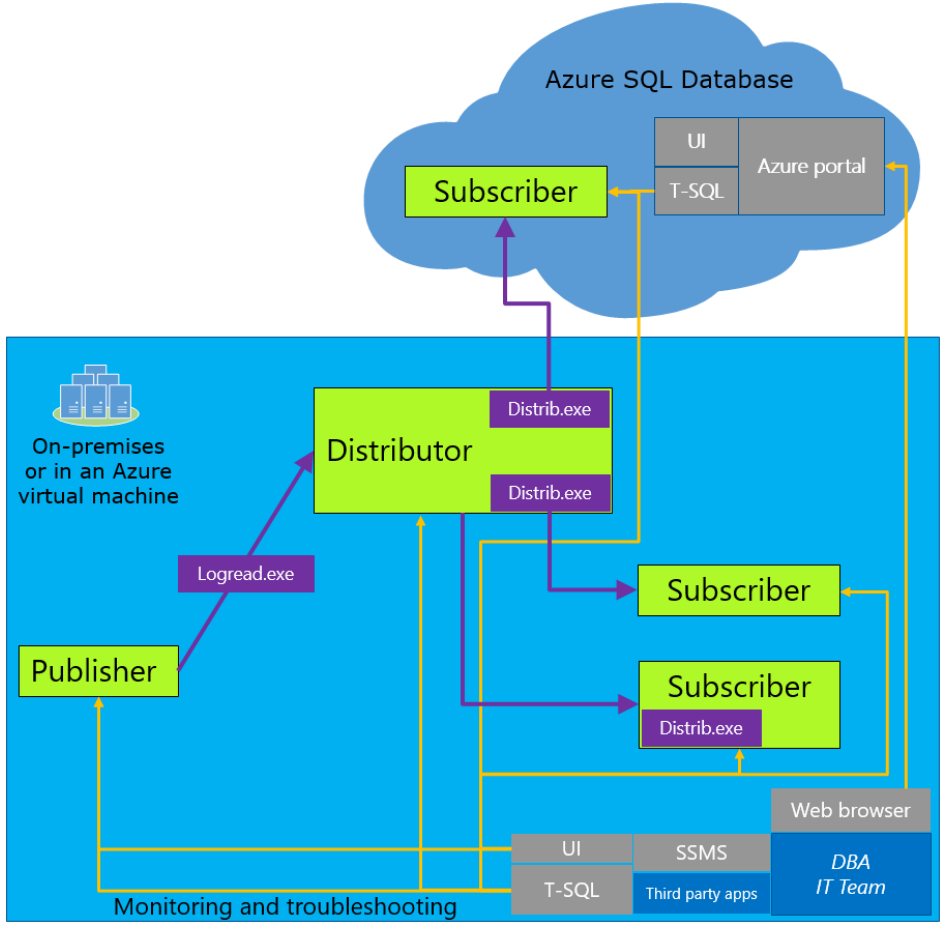
Method 3



Refactor with Visual Studio and Data Migration Assistant, final deploy with SSMS

Migration with minimal downtime

Transactional Replication



Data Migration Resources

Blog

https://blogs.msdn.microsoft.com/datamigration

Upgrade SQL Server

https://docs.microsoft.com/sql/database-engine/install-windows/upgrade-sql-server

Post-migration validation and optimization guide

https://docs.microsoft.com/sql/relational-databases/post-migration-validation-and-optimization-guide

Azure Database Migration Guide

https://datamigration.microsoft.com

Q&A